26 cm.

19 cm.

CETIZAL 5 mg Tablet

ANTIHISTAMINE FOR SYSTEMIC USE

Each film coated tablet contains:
Levocetirizine dihydrochloride 5 mg

PRODUCT DESCRIPTION:
White, oblong, biconvex, film coated tablet with engraved LC on one side and 5 on the left of the scored side

MECHANISM OF ACTION:
Pharmacotherapeutic group: Antihistamine for systemic use, piperazine derivative

Pharmacology
Levocetirizine, the (R) enantiomer of Cetirizine, is a potent and selective antagonist of peripheral H_1-receptors. Levocetirizine has high affinity for human H_1-receptors. Levocetirizine dissociates from H_1-receptors with a half-life of 115 ± 38 min. After single administration, Levocetirizine shows a receptor occupancy of 90% at 4 hours and 57% at 24 hours.

Pharmacokinetics
The pharmacokinetics of Levocetirizine are linear with dose- and time-independent with low inter-subject variability. The pharmacokinetic profile is the same when given as the single enantiomer or when given as Cetirizine. No chiral inversion occurs during the process of absorption and elimination.

Absorption
Levocetirizine is rapidly and extensively absorbed following oral administration. In adults, peak plasma concentrations are achieved 0.9 hour after dosing. The extent of absorption is dose-independent and is not altered by food, but the peak concentration is reduced and delayed.

Distribution
No tissue distribution data are available in humans, neither concerning the passage of Levocetirizine through the blood-brain-barrier.
Levocetirizine is 90% bound to plasma proteins. The distribution of Levocetirizine is restrictive, as the volume of distribution is 0.4 L/Kg.

Metabolism
The extent of metabolism of Levocetirizine in humans is less than 14% of the dose and therefore differences resulting from genetic polymorphism or concomitant intake of enzyme inhibitors are expected to be negligible. Metabolic pathways include aromatic oxidation, N- and O- dealkylation and taurine conjugation. Dealkylation pathways are primarily mediated by CYP 3A4 while aromatic oxidation involved multiple and/or unidentified CYP isoforms. Levocetirizine had no effect on the activities of CYP isoenzymes 1A2, 2C9, 2C19, 2D6, 2E1 and 3A4 at concentrations well above peak concentrations achieved following a 5 mg oral dose.
Due to its low metabolism and absence of metabolic inhibition potential, the interaction of Levocetirizine with other substances, or vice-versa, is unlikely.

Elimination
The plasma half-life in adults is 7.9 ± 1.9 hours. The half-life is shorter in small children.
The major route of excretion of Levocetirizine and metabolites is via urine, accounting for a mean of 85.4% of the dose. Excretion via feces accounts for only 12.9% of the dose. Levocetirizine is excreted both by glomerular filtration and active tubular secretion.

Special population

Renal impairment
The apparent body clearance of Levocetirizine is correlated to the creatinine clearance. It is therefore recommended to adjust the dosing intervals of Levocetirizine, based on creatinine clearance in patients with moderate and severe renal impairment. In end-stage renal disease subjects, the total body clearance is decreased by approximately 80% when compared to normal subjects.

Pediatric population
In children 6 to 11 years of age, C_{max} and AUC volumes were about 2 fold greater than in adults. Total body clearance was 30% greater and elimination half-life was 24% shorter when compared with adults.

Elderly
The disposition of racemic Cetirizine is dependent on renal function rather than on age. This finding would also be applicable for Levocetirizine, as Levocetirizine and Cetirizine are both predominantly excreted in urine. Therefore, the Levocetirizine dose should be adjusted in accordance with renal function in elderly patients.

Gender
Half-life was slightly shorter in women than in men. However, the body-weight adjusted oral clearance in women appeared to be comparable with that in men. The same daily dosage and dosing intervals are applicable for men and women with normal renal function.

Race
The effect of race on Levocetirizine has not been studied. As Levocetirizine is primarily renally excreted, and there are no important racial differences in creatinine clearance, pharmacokinetic characteristics of Levocetirizine are not expected to be different across races. No race-related differences in the kinetics of racemic Cetirizine have been observed.

Hepatic impairment
The pharmacokinetics of Levocetirizine in hepatically impaired subjects have not been tested. Patients with chronic liver diseases (hepatocellular, cholestatic, and biliary cirrhosis) given 10 or 20 mg of the racemic compound Cetirizine as a single dose had a 50% increase in half life along with a 40% decrease in clearance compared to healthy subjects.

Pharmacokinetic / pharmacodynamic relationship
The action on histamine-induced skin reactions is out of phase with the plasma concentrations.

INDICATION[2]:
Levocetirizine is indicated for the symptomatic treatment of allergic rhinitis (including persistent allergic rhinitis) and chronic idiopathic urticaria

DOSAGE AND ADMINISTRATION:
Oral

Posology

Adults and adolescents 12 years and above
The daily recommended dose is 5 mg once daily (one film coated tablet)[1].

Elderly
Adjustment of the dose is recommended in elderly patients with moderate to severe renal impairment (see Renal impairment below).

Renal impairment
The dosing intervals must be individualized according to renal function. Refer to the following table and adjust the dose as indicated. To use this dosing table, an estimate of the patient's creatinine clearance (CL_{cr}) in mL/min is needed. The CL_{cr} (mL/min) may be estimated from serum creatinine (mg/dL) determination using the following formula:

$$CL_{cr} = \frac{[140 - \text{age (years)}] \times \text{weight (Kg)}}{72 \times \text{serum creatinine (mg/dL)}} \quad (\times\ 0.85 \text{ for women})$$

Dosing adjustments for adult patients with impaired renal function:

Group	Creatinine clearance (mL/min)	Dosage and frequency
Normal	≥ 80	1 tablet once daily
Mild	50 - 79	1 tablet once daily
Moderate	30 - 49	1 tablet once every two days
Severe	< 30	1 tablet once every three days
End-stage renal disease - Patients undergoing dialysis	< 10	Contraindicated

In pediatric patients suffering from renal impairment, the dose will have to be adjusted on an individual basis taking into account the renal clearance of the patient and his body weight. There are no specific data for children with renal impairment.

Hepatic impairment
No dose adjustment is needed in patients with solely hepatic impairment. In patients with hepatic impairment and renal impairment, adjustment of the dose is recommended (see Renal impairment above).

Pediatric population
Children aged 6 to 12 years
The daily recommended dose is 5 mg once daily (one film coated tablet).
Children aged 2 to 6 years
No adjusted dosage is possible with the film coated tablet formulation. It is recommended to use a pediatric formulation of Levocetirizine.

Method of administration
The film coated tablet must be taken orally, swallowed whole with liquid and may be taken with or without food. It is recommended to take the daily dose in one single intake.

Duration of use
Intermittent allergic rhinitis (symptoms experienced for less than four days a week or for less than four weeks a year) has to be treated according to the disease and its history; it can be stopped once the symptoms have disappeared and can be restarted again when symptoms reappear. In case of persistent allergic rhinitis (symptoms experienced for more than four days a week or for more than four weeks a year), continuous therapy can be proposed to the patient during the period of exposure to allergens.
There is clinical experience with the use of Levocetirizine for treatment periods of at least 6 months. In chronic urticaria and chronic allergic rhinitis, there is a clinical experience of use of Cetirizine (racemate) for up to one year.

WARNING AND PRECAUTION:
Alcohol
Precaution is recommended with concurrent intake of alcohol (see Drug Interaction).

Risk of urinary retention
Caution should be taken in patients with predisposing factors of urinary retention (e.g. spinal cord lesion, prostatic hyperplasia) as Levocetirizine may increase the risk of urinary retention.

Risk of seizure aggravation
Caution should be taken in patients with epilepsy and patients at risk of convulsion as Levocetirizine may cause seizure aggravation.

Allergy skin tests
Response to allergy skin tests are inhibited by antihistamines and a wash out period (of 3 days) is required before performing them.

Excipients
Lactose
This product contains lactose. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

Withdrawal syndrome
Pruritus may occur when Levocetirizine is stopped even if those symptoms were not present before treatment initiation. The symptoms may resolve spontaneously. In some cases, the symptoms may be intense and may require treatment to be restarted. The symptoms should be resolved when the treatment is started.

Pediatric population
The use of the film coated tablet formulation is not recommended in children aged less than 6 years since this formulation does not allow for appropriate dose adaptation. It is recommended to use a pediatric formulation of Levocetirizine.

Warning[1]
This preparation contains Sodium metabisulfite that may cause serious allergic type reactions in certain susceptible patients. Do not use if known to be hypersensitive to bisulfites.

PREGNANCY AND LACTATION:
Fertility
There are no relevant data available.

Pregnancy
Caution should be exercised when prescribing to pregnant women. For Levocetirizine, no clinical data on exposed pregnancies are available. Animal studies do not indicate direct or indirect harmful effects with respect to pregnancy, embryonal/ fetal development, parturition or postnatal development.

Lactation
Caution should be exercised when prescribing to lactating women. Cetirizine is excreted in human breast milk. Levocetirizine is also expected to be excreted in human milk.

SIDE EFFECT[2]:
Adverse drug reactions (ADRs) are listed below by system organ class:

Nervous system disorders
Common: Headache, somnolence

Gastrointestinal disorders
Common: Dry mouth
Uncommon: Abdominal pain

General disorders and administration site conditions
Common: Fatigue
Uncommon: Asthenia

Post Marketing Data

Immune system disorders
Not known: Hypersensitivity including anaphylaxis

Metabolism and nutrition disorders
Not known: Increased weight, increased appetite

Psychiatric disorders
Not known: Aggression, agitation, hallucination, depression, insomnia, suicidal ideation, nightmares

Nervous system disorders
Not known: Convulsions, paraesthesia, dizziness, syncope, tremor, dysgeusia

Eyes disorders
Not known: Visual disturbances, blurred vision, oculogyration

Ear and labyrinth disorders
Not known: Vertigo

Cardiac disorders
Not known: Palpitations, tachycardia

Respiratory, thoracic, and mediastinal disorders
Not known: Dyspnea

Gastrointestinal disorders
Not known: Nausea, vomiting, diarrhea

Hepatobiliary disorders
Not known: Hepatitis, abnormal liver function test

Skin and subcutaneous tissue disorders
Not known: Angioneurotic edema, fixed drug eruption, pruritus, rash, urticaria

Musculoskeletal and connective tissues disorders
Not known: Myalgia, arthralgia

Renal and urinary disorders
Not known: Dysuria, urinary retention

General disorders and administration site conditions
Not known: Edema

Skin reactions occurring after discontinuation of Levocetirizine
After Levocetirizine discontinuation, pruritus has been reported (see Section Warning and Precaution).

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES:
Some patients could experience somnolence, fatigue and asthenia under therapy with Levocetirizine. Therefore, patients intending to drive, engage in potentially hazardous activities or operate machinery should take their response to the medicinal product into account.

CONTRAINDICATION:
Hypersensitivity to the Levocetirizine, to Cetirizine, to Hydroxyzine, to any other piperazine or to any of the other excipients.
Severe renal impairment at less than 10 mL/min creatinine clearance.

DRUG INTERACTION[2]:
No interaction studies have been performed with Levocetirizine (including no studies with CYP3A4 inducers); studies with the racemate compound Cetirizine demonstrated that there were no clinically relevant adverse interactions (with antipyrine, Pseudoephedrine, Cimetidine, Ketoconazole, Erythromycin, Azithromycin, Glipizide and Diazepam).

Theophylline
A small decrease in the clearance of Cetirizine can be observed with Theophylline (400 mg once a day); while the disposition of Theophylline cannot be altered by concomitant Cetirizine administration.

Ritonavir
The extent of exposure to Cetirizine is possible to increase while the disposition of Ritonavir can be slightly altered further to concomitant Cetirizine administration.

Food
The extent of absorption of Levocetirizine is not reduced with food, although the rate of absorption is decreased.

Alcohol
The concurrent administration of Cetirizine or Levocetirizine and alcohol or other CNS depressants may cause additional reductions in alertness and impairment of performance.

OVERDOSE AND TREATMENT:
Symptoms
Symptoms of overdose may include drowsiness in adults. In children agitation and restlessness may initially occur, followed by drowsiness.

Management of overdoses
There is no known specific antidote to Levocetirizine.
Should overdose occur, symptomatic or supportive treatment is recommended. Gastric lavage should be considered shortly after ingestion of the drug.
Levocetirizine is not effectively removed by hemodialysis.

STORAGE:
Store at temperature of not more than 30°C.

DOSAGE FORM AND PACKAGING AVAILABLE:
5 mg Tablet, Blister 10x10's

DATE OF REVISION:
May 10, 2023

Manufactured by:
UNISON LABORATORIES CO., LTD.
39 Moo 4, Klong Udomcholjorn, Muang Chachoengsao,
Chachoengsao 24000 Thailand

C-MY/100523-04 (AR)
(www.emc-medicines.org.uk, [1]Labeling Req't; [2]DCA-reg. Xyzal; [3]DCA 1st cor.)

ICMY 0081

PM Specification	Product	Code No.	Dimension	Packaging Type	Thickness
	CETIZAL 5	ICMY 0081	W 26 x L 19 cm.	Wood Free Paper (กระดาษปอนด์)	60 g (0.08 mm.)

Designed by: (PDM, ASPD, PDC-M, PDC-A, PDO-M, PDO-A)
Checked by: (PDM/ASPD)
Approved by: (MD (Only Pattern and Colors))
Approved by: PLC: (Date)
LRA: (Date)
IRA: (Date)
QCC-PM: (Date)
CUSTOMER/SALES DEPT.: (Date)

CETIZAL 5_Insert_ICMY 0081 By...DA 12/05/2023